

Matplotlib & Seaborn

↓
for visualisation

→ statistical plots
↓

```
avg_tip_by_day = data.groupby("day")["tip"].mean()
```

Thurs	2.77
Fri	2.73
Sat	2.99
Sun	3.25

Distributions
Categorical
comparisons
regression

→ customize the size of the plots

```
plt.figure(figsize=(6, 4))  
plt.plot(avg_tip_by_day.index, avg_tip_by_day.values, marker='o')  
plt.title('Average tip by day')  
plt.xlabel('Day')  
plt.ylabel('Average tip')  
plt.show()
```

time series data, weather forecast data

①



Line plot (trend)

②

bar plot → going for categorical

based comparisons (x → Cate

y → cont
data)

③

Pie chart → Distribution

of any given
data

④ Histogram / Distplot → distribution
of any given data

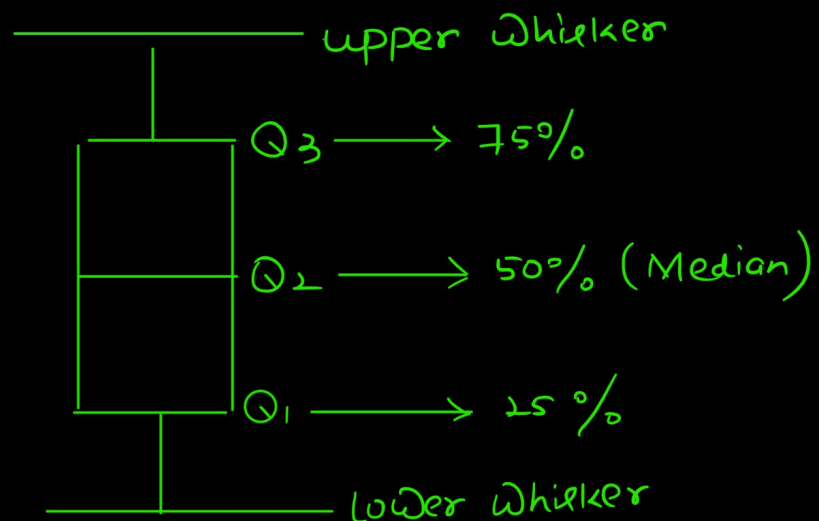
⑤ Boxplot → to check the outliers in
a given data

{ 50000, 70000, 75000, 65000,

5000, 1000000 }

extreme value

Considered as an outlier



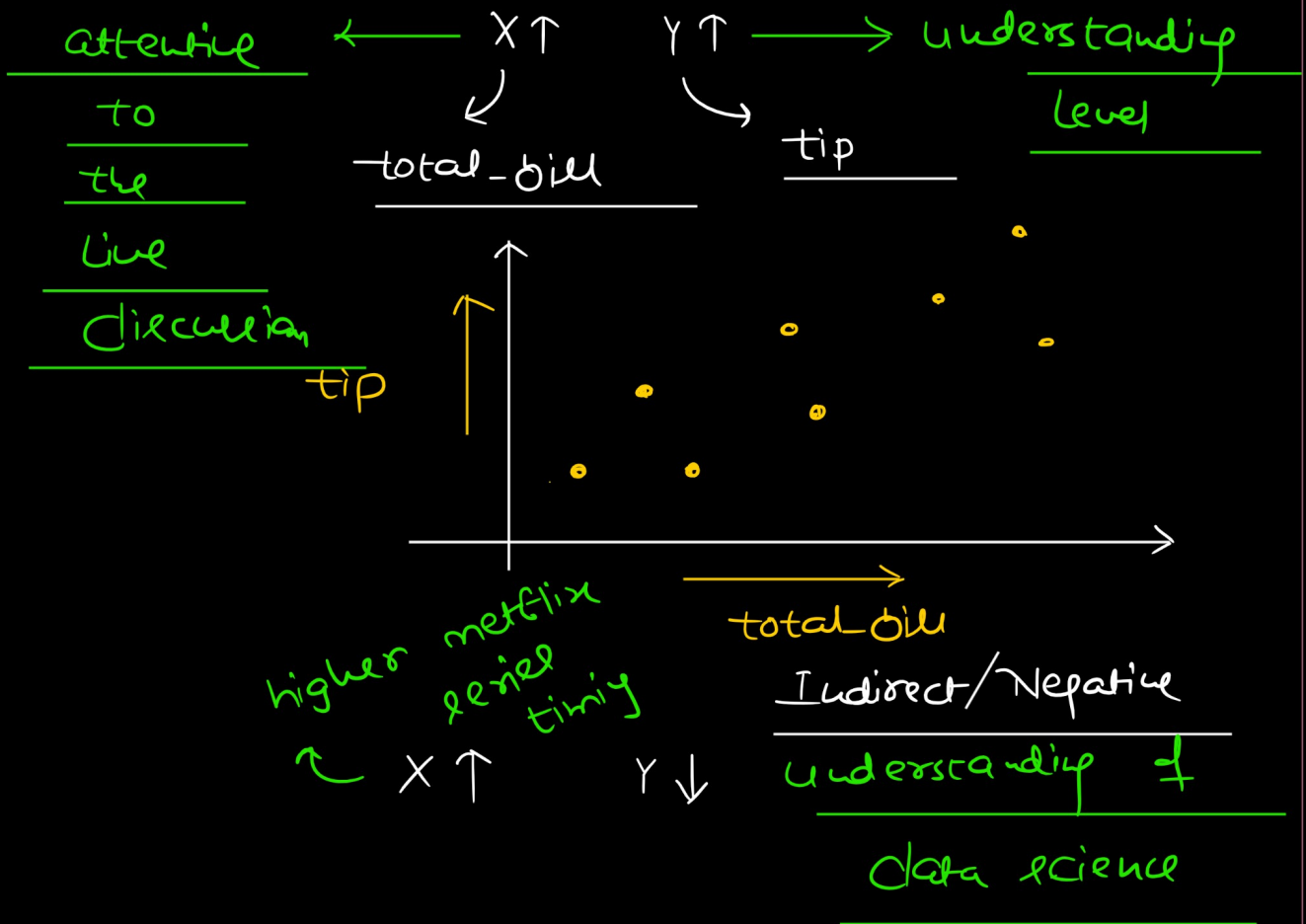
⑥ Relationship among two features

Scatterplot

Peplot

Correlation heatmap

Direct/Positive Relationship



7

Correlation heatmap (feature selection)

→
$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

to determine the strength of a given relationship

X ↑ Y ↑ → the relationship

X ↑ Y ↓ → -ve relationship

> 0.85 → strong correlation

(x, y) → similar value
interpretation to
predict the
final outcome

-1 to +1

-ve $\longrightarrow$ $x \uparrow$ $y \downarrow$

+ve $\longrightarrow$ $x \uparrow$ $y \uparrow$ (0.85)

x & y are
giving similar
info

>0.85 $\longrightarrow$ strongly correlation to
the model

to do

same prediction

~~$f_1 \longrightarrow$ DOB ✓~~
 $f_2 \longrightarrow$ Age } $\longrightarrow$ 0.90